Théo Alessandro Hermann

Roboticist and urban scientist · MIT Senseable City Lab Rio

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I'm Théo, a French-Brazilian roboticist and urban scientist. Paris, São Paulo, Rio, Lausanne, Amsterdam, Barcelona, Stockholm.

I trained in robotics at EPFL, built perception for a person-following robot at VITA Lab and for an autonomous boat at Roboat, then brought the same tools to cities at MIT Senseable City Lab.

Wherever I work, the question is the same: what aren't we seeing yet that explains how the system behaves? Today I help robot makers find out why their systems fail, and I study how city form shapes light and air.

EXPERIENCE

Research Fellow Jan 2025 – Present
MIT Senseable City Lab Rio · Rio de Janeiro, Brazil

There from the first week of the Rio branch. I work on morphometrics of informal urban form: describing favela fabric from terrain models and building footprints so that wind and sunlight simulation has something defensible to run on. Tech: Python, GeoPandas, point-cloud tooling.

Computer Vision and MLOps Intern (first intern) Mar 2024 – Aug 2024
Roboat (MIT Spinoff) · Amsterdam, Netherlands

Brought camera-based perception to a stack that until then ran on LiDAR: detection and multi-object tracking for autonomous canal vessels on Jetson Orin, and the MLOps around it, with DVC versioning over the on-water recordings the models were trained on. Tech: PyTorch, ROS 2, Docker, DVC.

Visiting Research Fellow Oct 2023 – Feb 2024
IRI (CSIC-UPC) · Barcelona, Spain

Built PyTorch fusion model combining neuromorphic event cameras with visual-inertial odometry for 6-DOF drone state estimation during aggressive maneuvers (>5g). Tech: PyTorch, ROS, C++.

Research Assistant Jun 2022 – Jun 2023
VITA Lab, EPFL · Lausanne, Switzerland

Redid the architecture and the perception block of the lab's ROS autonomy stack for mobile robot navigation in crowded environments, continuing the VITA Lab research line. Real-time detection, multi-object tracking, 3D pose estimation, and trajectory prediction on Segway Loomo. Tech: ROS, Python, C++, OpenCV.

EDUCATION

Master's degree (M.Sc), Robotics — EPFL 2021–2024
Thesis on urban tree canopy from aerial LiDAR, with Senseable City Lab, MIT and the AMS Institute.

Exchange (B.Sc 3rd year), Embedded Systems — KTH Royal Institute of Technology 2020–2021

Bachelor's degree (B.Sc), Microengineering — EPFL 2018–2021

TECHNICAL SKILLS

Robotics	ROS/ROS2, SLAM, path planning, sensor fusion, Jetson/edge deployment
ML / AI	PyTorch, scikit-learn, reinforcement learning
Perception	LiDAR processing, point clouds, stereo vision, event cameras, 3D reconstruction
Dev	Python, C++, Git, Docker, CI/CD, DVC, MLOps
Tools	OpenFOAM CFD, Blender, GeoPandas

LANGUAGES

French Native · **English** Fluent · **Portuguese** Fluent · **Spanish** Conversational